DO HOANG MINH HAN

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OBJECTIVE

I am an Electronics and Telecommunications Engineering undergraduate student with research experience in wireless communications, including physical-layer security, reconfigurable intelligent surfaces, and IoT. My short-term goal is to deepen my knowledge of wireless communications and strengthen my research expertise through collaborative projects. In the long term, I aspire to develop secure, energy-efficient next-generation wireless systems by integrating signal processing, machine learning, and optimization through analytical research and practical engineering.

EDUCATION

TON DUC THANG UNIVERSITY (TDTU) Ho Chi Minh City, Vietnam
B.Eng. in Electronics and Telecommunications Engineering (in progress) *September 2023–Present*
Current GPA: 8.26/10.00 | Current cohort rank: 1st. | Expected graduation: September 2027

WORK EXPERIENCE

Wireless Communications Research Group (WiCOM), *Research Intern* *September 2026 – Present*

- Investigate digital signal processing for wireless communications, focusing on modulation and demodulation techniques.
- Explore machine learning approaches and their applications in wireless communication systems.

Advanced Intelligent Technology Research Group (AITECH), *Undergraduate Researcher* *October 2025 – Present*

- Conduct analytical research on physical-layer security and RF energy harvesting in backscatter-enabled IoT networks.
- Co-developed a source-assisted backscatter framework under active full-duplex eavesdropping; derived outage and intercept probability expressions and validated the analysis through MATLAB-based Monte Carlo simulations.
- Contribute to collaborative studies through literature review, numerical analysis, and manuscript preparation and revision.

Ton Duc Thang University, *Student and Project Leadership* *2023 – Present*

- Served as secretary of the class Youth Union branch and executive committee member of the class Student Association (2023–2024; 2025–2026). In 2024–2025, served as vice-chair of the class Student Association, student group leader, and executive committee member of the Faculty Youth Union.
- Lead the team for the project *CNN-Based Cooperative Spectrum Sensing for Cognitive Radio Networks*; previously led the projects *Human-Following Robot with Fall Detection* and *Computer-Vision-Based Offside Detection in Football Video*.

PUBLICATIONS

A. Journal Articles

1. Performance Analysis of Multi-Antenna RIS-Assisted Nonlinear Energy Harvesting Systems over λ–κ–μ Shadowed Fading with Phase Noise *2026*
IEEE Transactions on Wireless Communications (In preparation)
Thien P. Nguyen, **Minh-Han H. Do**, Lam-Thanh Tu, Phuong T. Tran, and Kyungchun Lee

2. Security-Reliability Analysis of Backscatter Communications for Copyright-Protected Content Delivery in IoT Networks *2026*
Scientific Reports (Minor Revision)
Minh-Han H. Do, Hye-Young Kim, Thu-Ha Thi Pham, Tan N. Nguyen, Phuong T. Tran, Byung-Seo Kim, and Miroslav Voznak

3. Secrecy Performance of Energy Harvesting Backscatter-Assisted Hybrid Long and Short-Packet Communications *2026*
Advances in Electrical and Electronic Engineering (Accepted)
Nguyen Quang Sang, Tran Cong Hung, **Minh-Han H. Do**, Bui Vu Minh, Tran Hoang Quang Minh, and Rupak Kharel

4. Computational Analysis of Secrecy Outage in Energy-Harvesting Multi-Backscatter IoT Networks *2026*
Radioengineering (Online first: 30 September 2026)
Sang Q. Nguyen, **Minh-Han H. Do**, Lam-Dong Huynh, and Tan N. Nguyen

5. On the Performance of Power-Splitting SWIPT AF Relay-Assisted NOMA System with Imperfect SIC *2026*
Advances in Electrical and Electronic Engineering (Early Access)
Quang-Sy Vu, **Minh-Han H. Do**, Quang-Sang Nguyen, Dinh-Tung Vo, Duc Ngoc Minh Dang, and Rupak Kharel

B. Conference Papers

1. Joint SNR–EH Performance of Diversity Combining Techniques in LoRa Downlink Networks *2026*
2026 International Conference on Advanced Technologies for Communications (ATC 2026) (Accepted)
Thi-Phuong-Anh Hoang, **Minh-Han H. Do**, Thien Huynh-The, and Lam-Thanh Tu

SELECTED PROJECTS

CNN-Based Cooperative Spectrum Sensing for Cognitive Radio Networks, *Leader* *2026*

- Developing cooperative spectrum sensing with local energy detection at multiple secondary users and a fusion-center CNN inferring primary-user activity from distributed data. Planned evaluation covers detection and false-alarm probabilities and sensing error versus SNR, user count, and sample size, benchmarked against conventional sensing/fusion rules.

Human-Following Robot with Fall Detection, *Leader* *2026*

- Developed a real-time AI–IoT robot for person following and human-state monitoring using YOLO person/posture detection on Raspberry Pi, Arduino motor control, and serial communication. Integrated fall and person-loss detection, web monitoring and alerts, backend event handling, database storage, user authentication, and device/safety-status dashboards.

GPS/GSM-Based Motorcycle Tracking and Anti-Theft System, *Individual Embedded Systems Project* *2026*

- Built a PIC16F877A prototype integrating NEO-6M GPS, A7680C mobile communication, vibration sensing, and local alarm circuitry; developed C control software and designed the PCB. Implemented GPS coordinate processing, geofencing, last-known-position storage, and remote SMS warnings; practical tests verified position display, anti-theft mode activation, and warnings when leaving the predefined safe zone.

Computer-Vision-Based Offside Detection in Football Video, *Leader* *2026*

- Developed an OpenCV video pipeline for player localisation and frame-level candidate offside-line estimation using YOLO object detection, K-means team classification, and perspective geometry. Applied Canny edge detection, Hough transforms, RANSAC vanishing-point estimation, and Kalman filtering to identify pitch geometry and stabilise spatial estimates.

TECHNICAL SKILLS

- **Programming and tools:** MATLAB, Python, C/C++, and LaTeX.
- **Modeling and simulation:** Wireless channel modeling, outage/secrecy analysis, Monte Carlo simulation, and numerical validation.
- **Teamwork and leadership:** Research collaboration and student project team leadership.

RESEARCH DIRECTIONS

Physical-layer security and performance analysis of energy-efficient wireless networks, with an interest in machine learning for wireless communications.

HONORS & AWARDS

- **Academic merit scholarship, TDTU** *2026*
- **Second prize in MATLAB signal processing innovation, TDTU** *2026*

CERTIFICATIONS

- **Aptis ESOL B2**

REFERENCES

1. **Assoc. Prof. Phuong T. Tran, Ph.D.** (*Dean and Department Head*), Faculty of Electrical and Electronics Engineering, Ton Duc Thang University. E-mail: tranthanhphuong@tdtu.edu.vn. Google Scholar
2. **Assoc. Prof. Tan Nhat Nguyen, Ph.D.** (*Head of AITECH Research Group*), Department of Electronics and Telecommunications Engineering, Ton Duc Thang University. E-mail: nguyennhattan@tdtu.edu.vn. Google Scholar.
3. **Lam-Thanh Tu, Ph.D.** (*Lecturer*), Department of Electronics and Telecommunications Engineering, Ton Duc Thang University. E-mail: tulamthanh@tdtu.edu.vn. Google Scholar